

**N2 Normal distribution A
level only**

Name: _____

Class: _____

Date: _____

Time: **332 minutes**

Marks: **280 marks**

Comments:

EXAM PAPERS PRACTICE

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Q1.

- (a) The volume, X millilitres, of toothpaste in medium-sized tubes may be assumed to be normally distributed with a mean of 56 and a standard deviation of 2.5.

Determine the probability that the volume of toothpaste in a tube is:

- (i) less than 60 ml; (3)
- (ii) between 50 ml and 60 ml; (3)
- (iii) exactly 55 ml. (1)

- (b) The volume, Y millilitres, of toothpaste in large-sized tubes may be assumed to be normally distributed with a standard deviation of 3.4.

Given that 98 per cent of these tubes contain more than 100 ml of toothpaste, determine the mean volume of toothpaste in a large-sized tube.

(4)

(Total 11 marks)

Q2.

The weight, X grams, of a particular variety of orange is normally distributed with mean 205 and standard deviation 25.

- (a) Determine the probability that the weight of an orange is:
- (i) less than 250 grams; (3)
 - (ii) between 200 grams and 250 grams. (3)

- (b) A wholesaler decides to grade such oranges by weight. He decides that the smallest 30 per cent should be graded as small, the largest 20 per cent graded as large, and the remainder graded as medium.

Determine, to one decimal place, the maximum weight of an orange graded as:

- (i) small; (5)
- (ii) medium.

- (c) The weight, Y grams, of a second variety of orange is normally distributed with mean 175.

Given that 90 per cent of these oranges weigh less than 200 grams, calculate the

standard deviation of their weights.

(4)

(Total 15 marks)

Q3.

The length, X centimetres, of eels in a river may be assumed to be normally distributed with mean 48 and standard deviation 8.

An angler catches an eel from the river. Determine the probability that the length of the eel is:

(a) exactly 60 cm;

(1)

(b) less than 60 cm;

(2)

(c) within 5% of the mean length.

(5)

(Total 8 marks)

Q4.

The volume, L litres, of emulsion paint in a plastic tub may be assumed to be normally distributed with mean 10.25 and variance σ^2 .

(a) Assuming that $\sigma^2 = 0.04$, determine $P(L < 10)$

(4)

(b) Find the value of σ so that 98% of tubs contain more than 10 litres of emulsion paint.

(4)

(Total 8 marks)

Q5.

The weights of bags of red gravel may be modelled by a normal distribution with mean 25.8 kg and standard deviation 0.5 kg.

(a) Determine the probability that a randomly selected bag of red gravel will weigh:

(i) less than 25 kg;

(3)

(ii) between 25.5 kg and 26.5 kg.

(4)

(b) Determine, to two decimal places, the weight exceeded by 75% of bags.

(4)

(Total 11 marks)

Q6.

- (a) The time, X minutes, taken by Fred Fast to install a satellite dish may be assumed to be a normal random variable with mean 134 and standard deviation 16.

(i) Determine $P(X < 150)$.

(3)

(ii) Determine, to one decimal place, the time exceeded by 10 per cent of installations.

(4)

- (b) The time, Y minutes, taken by Sid Slow to install a satellite dish may also be assumed to be a normal random variable, but with

$$P(Y < 170) = 0.14 \quad \text{and} \quad P(Y > 200) = 0.03.$$

Determine, to the nearest minute, values for the mean and standard deviation of Y .

(6)

(Total 13 marks)

Q7.

Garden canes have lengths that are normally distributed with mean 208.5 cm and standard deviation 2.5 cm.

- (a) Show that the probability of the length of a randomly selected cane being between 205 cm and 210 cm is 0.645, correct to three decimal places.

(4)

- (b) Ten canes are selected at random.

Calculate the probability that exactly 6 of these canes have lengths between 205 cm and 210 cm.

(3)

(Total 7 marks)

Q8.

A steel rolling mill has two machines, A and B , for cutting steel bars. For each machine the length of a cut bar can be modelled by a normal distribution.

- (a) Bars cut by machine A have a mean length of 1212 mm and a standard deviation of 5 mm.

(i) Determine the probability that the length of a bar is greater than 1205 mm.

(4)

(ii) Calculate the length exceeded, on average, by one bar in five hundred.

(5)

Bars with lengths less than 1200 mm are rejected as too short, and bars with lengths greater than 1225 mm are rejected as too long.

- (b) Of the bars cut by machine B , 1.5 per cent are rejected as too short and 1.9 per cent are rejected as too long. Calculate, to one decimal place, the mean and standard deviation of the lengths of the bars cut by this machine.

(7)

(Total 16 marks)

Q9.

The distance, in kilometres, travelled to work by the employees of a city council may be modelled by a normal distribution with mean 7.5 and standard deviation 2.5.

- (a) Find the probability that the distance travelled to work by a randomly selected employee of the city council is:

(i) less than 11.0 km;

(3)

(ii) between 5.5 km and 10.5 km.

(4)

- (b) Find d such that 10% of the council's employees travel less than d kilometres to work.

(4)

(Total 11 marks)

Q10.

Jars of bolognese sauce, sold by a supermarket, are stated to have contents of weight 500 g. The weights, in grams, of the actual contents of jars in a large batch are normally distributed with mean 506 and standard deviation 5.

- (a) Find the probability that the contents of a randomly selected jar will weigh:

(i) less than 500 g;

(3)

(ii) between 495 g and 505 g.

(4)

- (b) Find the weight which is exceeded by the contents of 99.9% of the jars in this batch.

(4)

(Total 11 marks)

Q11.

In an office a machine dispenses hot water for use in making tea, coffee or hot chocolate. The machine is filled with water at the beginning of each day. When 20 litres of water have been used the machine must be refilled.

- (a) The daily demand for hot water from the machine may be modelled by a normal distribution with mean 26 litres and standard deviation 8 litres. Find the probability that, on a particular day, the machine:

- (i) will not need to be refilled;
 - (ii) will need to be refilled exactly once.
- (5)**
- (b) Find the probability that during a five-day week the demand for hot water from the machine will be less than 20 litres on each day;
- (2)**
- (c) The machine is moved to a different office. Following the move, it is observed that the machine needs refilling at least once on 84% of days and needs refilling more than once on 12% of days. Assuming that the daily demand for hot water, following the move, may still be modelled by a normal distribution, find the mean and standard deviation.
- (6)**
- (Total 13 marks)**

Q12.

The heights of female students attending a sixth form college have a mean of 168.0 cm and a standard deviation of 4.5 cm. The heights can be modelled by a normal distribution.

Find the probability that the height of a randomly selected female student attending this college is:

- (a) less than 172.5 cm;
- (b) between 159 cm and 163.5 cm.

(Total 7 marks)

Q13.

A coin-operated machine dispenses soft drinks into cups when a button is pressed.

- (a) After the machine has been overhauled the amount, X millilitres, of soft drink dispensed on each press of the button may be modelled by a normal distribution with mean 490 and standard deviation 10.

- (i) Find the probability that X lies between 475 and 510.
- (5)**

- (ii) Find the value of X which will be exceeded on 97% of presses of the button.
- (4)**

- (b) The owner of the machine receives complaints that some cups are underfilled and suspects that the machine is due for an overhaul. Investigation shows that the amount, Y millilitres, of soft drink dispensed is less than 475 on 15% of presses of the button.

Find the mean of Y , assuming that it may be modelled by a normal distribution with standard deviation 10.

(4)
(Total 13 marks)

Q14.

Each match in a 7-a-side football competition is scheduled to take 15 minutes (seven minutes each way plus one minute for half-time). However, due to delays, the actual time taken by a match may be modelled by a normal distribution with mean 19.5 minutes and standard deviation 1.2 minutes.

- (a) Find the probability that a match will last between 16.5 and 18.0 minutes.

(5)

- (b) For future competitions the organisers request that referees reduce delays to a minimum, so that matches finish on schedule. Following this request, it is observed that 80% of matches last less than 18 minutes and only 4% of matches last more than 19 minutes. Assuming that the lengths of matches may still be modelled by a normal distribution, find the mean and standard deviation.

(6)

(Total 11 marks)

Q15.

An airline operates a service between Manchester and Paris. The flight time may be modelled by a normal distribution with mean 85 minutes and standard deviation 8 minutes.

- (a) Find the probability that a particular flight time is:

- (i) less than 75 minutes;

(3)

- (ii) between 75 minutes and 81 minutes.

(3)

- (b) In order to gain publicity for the service, the airline decides to refund fares when a flight time exceeds q minutes. Find the value of q such that the probability of fares being refunded on a particular flight is 0.001.

(4)

- (c) Find the probability that the mean of four randomly selected flight times will be less than 81 minutes.

(4)

(Total 14 marks)

Q16.

Kevin uses his mobile phone for X minutes each day. X is a random variable which may be modelled by a normal distribution with mean 28 minutes and standard deviation 8 minutes.

- (a) Find the probability that on a particular day Kevin uses his mobile phone for:

- (i) less than 30 minutes;

(4)

- (ii) between 10 and 20 minutes.

(4)

- (b) Calculate an interval, symmetrical about 28 minutes, within which $\bar{X}$ will lie on 80% of days.

(4)

(Total 12 marks)

Q17.

The weights, in grams, of the contents of tins of mackerel fillets are normally distributed with mean μ and standard deviation 2.5. The value of μ may be adjusted as required.

- (a) Find the proportion of tins with contents weighing between 125.0 grams and 130.0 grams when $\mu = 129.0$.

(5)

- (b) (i) State, without proof, the value of μ which would maximise the proportion of tins with contents weighing between 125.0 grams and 130.0 grams.

(1)

- (ii) Find the proportion of tins with contents weighing between 125.0 grams and 130.0 grams when μ is equal to the value you have specified in part (b)(i).

(3)

- (c) Find, to one decimal place, the value of μ such that 99% of the tins have contents weighing more than 125.0 grams.

(4)

(Total 13 marks)

Q18.

The time taken for a bus to complete its journey from Sunderland city centre to its terminus may be modelled by a normal distribution with mean 34 minutes and standard deviation 5 minutes.

- (a) Find the probability that the bus will take between 30 and 40 minutes to complete its journey.

(5)

- (b) Find the length of time which will be exceeded by 90% of these bus journeys.

(4)

(Total 9 marks)

Q19.

Rahul works for a bank. He can travel to the branch where he works by car, bus or bicycle. He does not wish to arrive too early as he cannot enter the building before it is unlocked at 8.30 am. He does not wish to arrive after his starting time, which is 9.00 am.

- (a) When he travels by bus his journey time may be modelled by a normally distributed random variable with mean 40 minutes and standard deviation 10 minutes. Given that he leaves home at 8.00 am and travels by bus, find the probability that he will

arrive at the branch:

- (i) before 9.00 am;
 - (ii) between 8.30 am and 9.00 am. (5)
- (b) When he travels by bicycle his journey time may be modelled by a normally distributed random variable with mean 35 minutes and standard deviation 2 minutes. What time should he leave home when he travels by bicycle if he wishes to have the probability of 0.99 of arriving before 9.00 am? (5)
- (c) When he travels by car his journey time may be modelled by a normally distributed random variable with mean 28 minutes and standard deviation 8 minutes. Explain briefly why, when he travels by car, he should leave home at 8.17 am in order to maximise his chance of arriving between 8.30 am and 9.00 am. (2)
- (d) Give an advantage of:
- (i) travelling by car compared to bus or bicycle; (1)
 - (ii) travelling by bicycle compared to bus or car. (1)

Your answers should be based only on the data given in the question and not on general considerations such as exercise, environmental consequences and comfort.

(Total 14 marks)

Q20.

A gas supplier maintains a team of engineers who are available to deal with leaks reported by customers. Most reported leaks can be dealt with quickly but some require a long time. The time (excluding travelling time) taken to deal with reported leaks is found to have a mean of 65 minutes and a standard deviation of 60 minutes.

- (a) Assuming that the times may be modelled by a normal distribution, estimate the probability that:
- (i) it will take more than 185 minutes to deal with a reported leak; (3)
 - (ii) it will take between 50 minutes and 125 minutes to deal with a reported leak; (4)
- (b) A statistician, consulted by the gas supplier, stated that, as the times had a mean of 65 minutes and a standard deviation of 60 minutes, the normal distribution would not provide an adequate model.

Explain the reason for the statistician's statement.

(2)

(Total 9 marks)

Q21.

Charlotte is a dentist. The time that each patient, attending for an appointment, spends in her surgery may be modelled by a normal distribution with mean 23 minutes and standard deviation 4 minutes.

(a) Find the probability that a patient attending for an appointment will spend:

(i) less than 28 minutes in her surgery;

(3)

(ii) between 20 and 30 minutes in her surgery.

(4)

(b) Find the length of time spent in her surgery which will be exceeded by 1% of patients.

(4)

(c) Calculate limits, symmetrical about the mean, within which 90% of times spent by patients in her surgery will lie.

(4)

(Total 15 marks)

Q22.

The number of miles that Anita's motorbike will travel on one gallon of petrol may be modelled by a normal distribution with mean 135 and standard deviation 12.

(a) Given that Anita starts a journey with one gallon of petrol in her motorbike's tank, find the probability that, without refuelling, she can travel:

(i) more than 111 miles;

(3)

(ii) between 141 and 150 miles.

(4)

(b) Find the longest journey that Anita can undertake, if she is to have a probability of at least 0.9 of completing it on one gallon of petrol.

(4)

(Total 11 marks)

Q23.

A meter is used to monitor the amount of dust in a factory atmosphere. A large number of readings was taken when the atmosphere was satisfactory and it was found that these readings could be modelled by a normal distribution with mean 266 and standard deviation 32.

(a) Find the probability that, when the atmosphere is satisfactory, the meter will give a reading between 250 and 300.

(5)

(b) The factory Health and Safety Committee recommends that action be taken to purify

the atmosphere if the machine gives a reading greater than k . The value of k is chosen so that the probability of the meter giving a reading greater than k , when the atmosphere is satisfactory, is 0.04.

Find the value of k .

(4)

(Total 9 marks)

Q24.

The lengths of components produced by a particular machine may be modelled by a normal distribution with mean 901.0 mm and standard deviation 2.0 mm. Find the probability that the length of a randomly selected component will be less than 902.5 mm.

(Total 4 marks)

Q25.

A study showed that the time, T in minutes, spent by a customer between entering and leaving Fely's department store has a mean of 20 with a standard deviation of 6.

Assume that T may be modelled by a normal distribution.

- (a) Find the value of T exceeded by 20% of customers.

(4)

- (b) (i) Write down the standard deviation of the mean time spent in Fely's store by a random sample of 90 customers.

(1)

- (ii) Find the probability that this mean time will exceed 21 minutes.

(4)

(Total 9 marks)

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